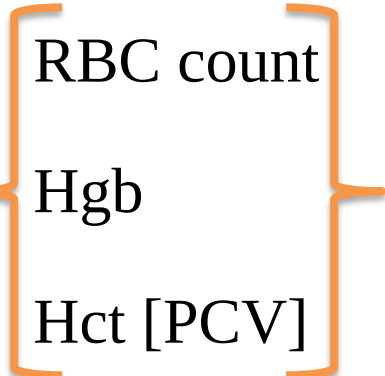


□ Blood and Immunity

□ Manual

1. TWBC count
 2. Differential WBC count [N-L-M-E-B]
 3. RBC count
 4. Hgb
 5. Hct [PCV]
 6. RI [MCV-MCH-MCHC]
 7. ESR
- 

1.Total White Blood Cells Count [TWBC]=[T.Leukocytes]

☐ Significance of the test

- ☐ To diagnose leukemia and leucopenia

- ☐ Help to diagnose disorders related to having a high WBC count or low WBC count.

 - ✓ High WBCs [Autoimmune & inflammatory disease]

 - ✓ Low WBCs [HIV]

- ☐ To investigate infections and unexplained fever

- ☐ To monitor treatments

Principle of the test

- ❑ **Whole blood** is diluted 1:20 in weak **Acetic Acid** solution which lyses red blood cells [RBCs] but preserves WBC.
- ❑ After WBCs counted
 - ✓ The count is multiplied by **Dilution Factor [DF]** and **Volume Correction Factor [VCF]**.
- ❑ The number of WBC is reported as number of cells per microliter (μl) of whole blood.

Specimen

- ☐ EDTA anticoagulated blood or
- ☐ Capillary blood can be used for counting WBCs.

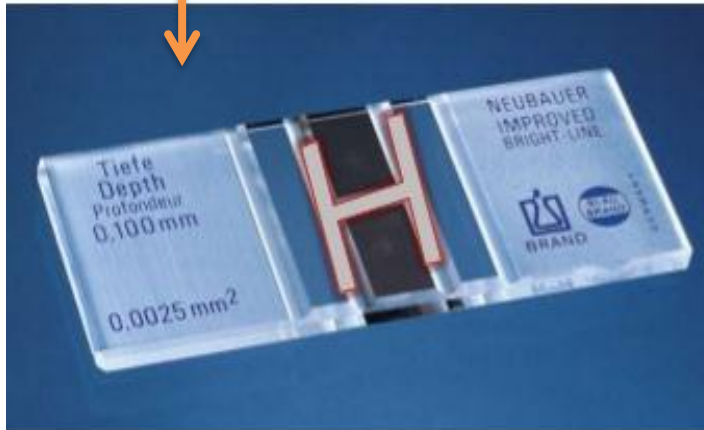
Materials and Reagents

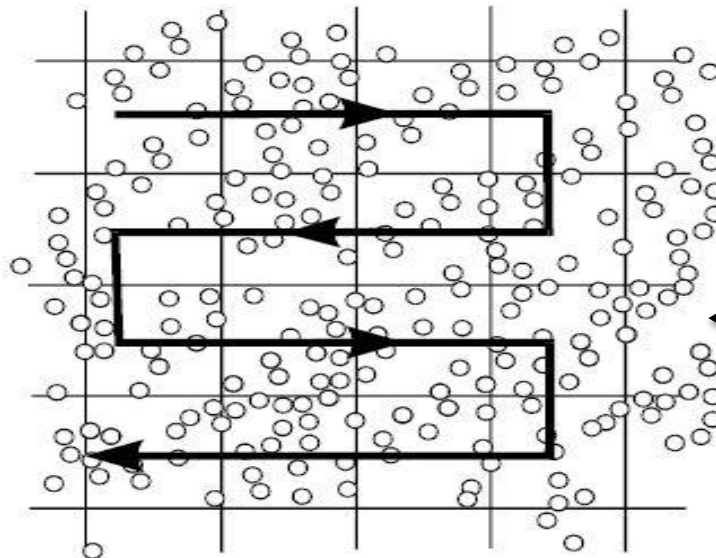
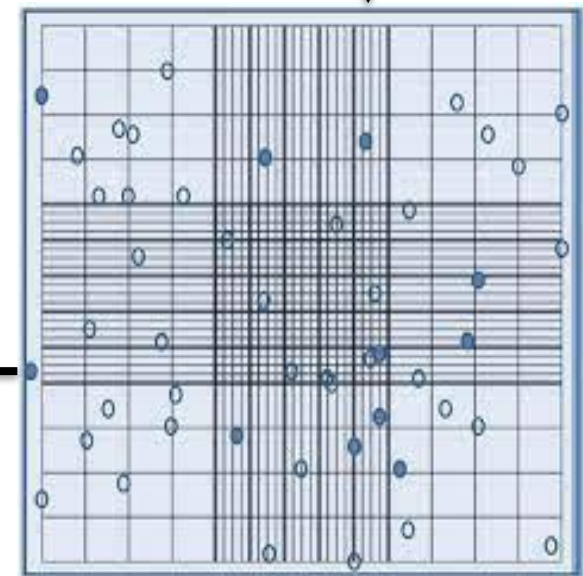
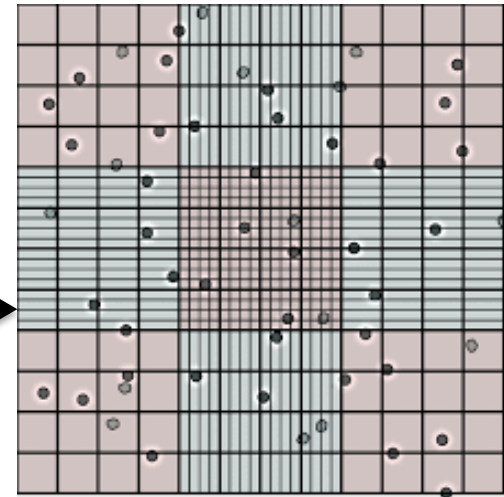
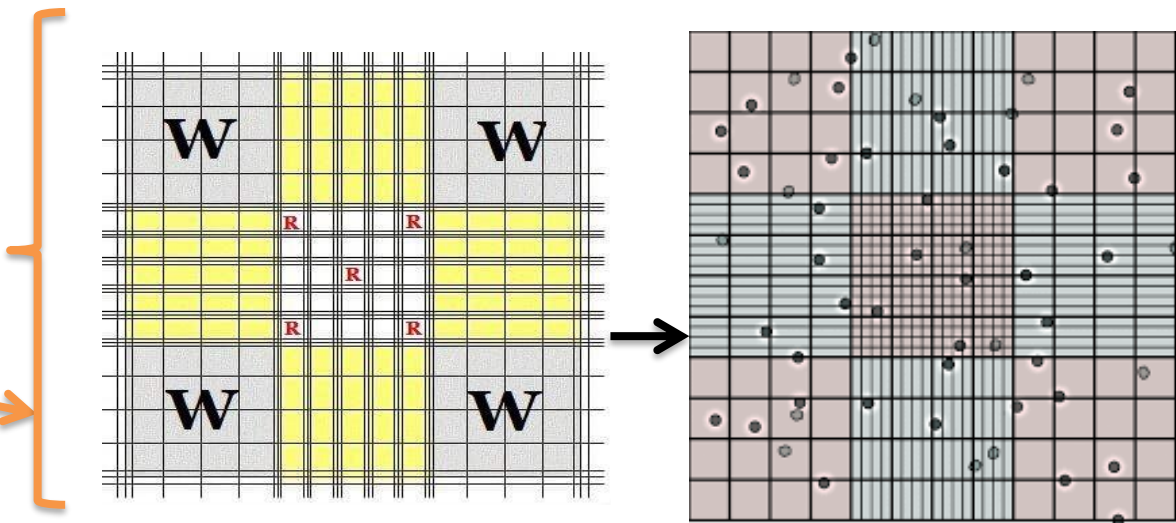
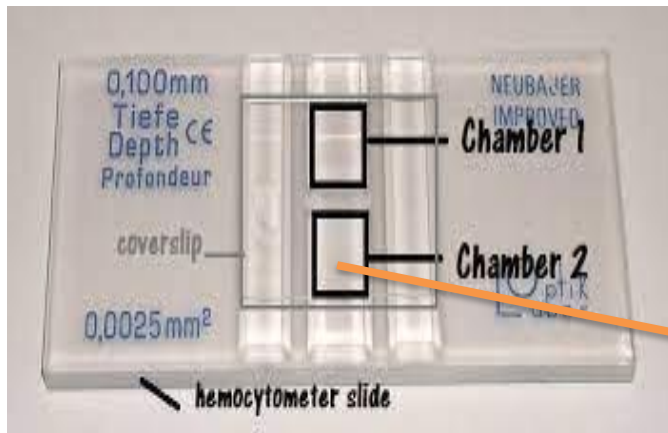
- ☐ **Hemocytometer (Improved Neubauer Chamber).**
- ☐ WBC Thoma Pippet → → Mark 11
- ☐ Diluting fluid [**Weak Acid** = Acetic Acid = Diluting fluid]
- ☐ Cover Glass
- ☐ Cotton
- ☐ Microscope
- ☐ Alcohol

Cont'd

Haemocytometer (Manual method)

Improved Neubauer Chamber





WBC Thoma Pippet



Short stem

bead

bulb

Long
stem

-WBC bulb is smaller than RBC

-Upper calibration = 11

-Constant volume of pipet = $11 - 1 = 10$

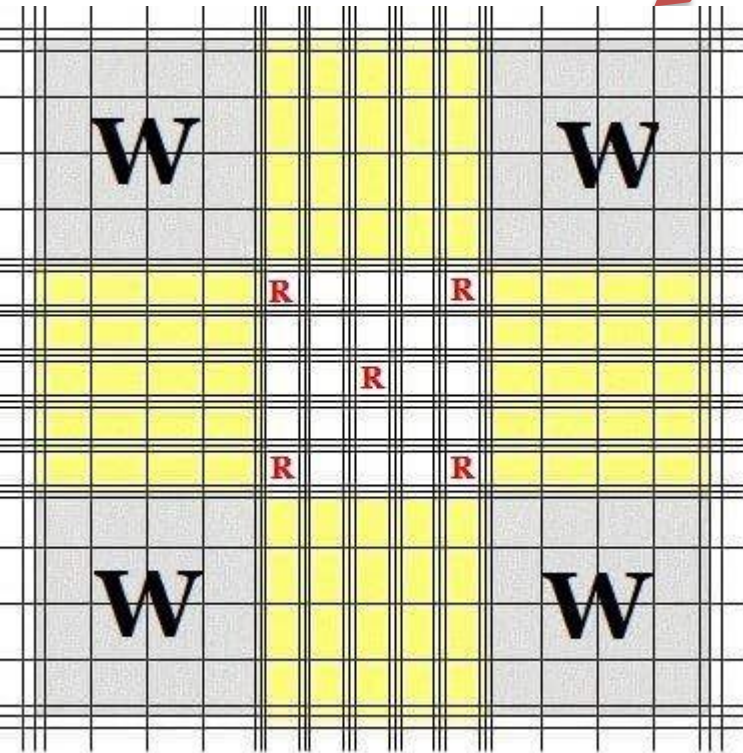
RBC - Volume of bulb is 10 times greater
than volume of bulb Of WBC.

Procedure

1. Mix the anticoagulated blood carefully
2. Suck the blood into pipette exactly up to **0.5** mark.
3. Wipe the excess blood outside the pipette using cotton
4. Take **diluting fluid** up to **11** mark
5. Gently rotate the pipette for mixing
6. Discard the first drops of **unmixed solution** present at the stem of the pipette
7. Charge the counting chamber and allow 2 minutes for settling of cells
8. Count WBCs in the 4 corner squares of the haemocytometer using microscope.

About the Haemocytometer

$$\underline{1\mu\text{L}=\text{mm}^3}$$



- ☐ The chamber has **9** large squares
- ☐ Each square is 1mm by 1mm (area = 1mm^2) and 0.1mm depth (volume = 0.1mm^3)
- ☐ WBC is count in the **4** large Corner squares
- ☐ Volume used for counting WBC is $4 \times 0.1 = \underline{0.4\text{ mm}^3}$.

Calculation

$$\text{\#WBC Count} = N_{\underline{o}} \text{ of cells counted} \times \text{DCF} \times \text{VCF}$$

$$\text{\#DCF} = 20 \text{ (1:20)}$$

$$\begin{aligned} \text{\#VCF} &= \text{Volume desired} / \text{Volume used} \\ &= 1 \text{ cumm} / 0.4 \text{ cumm} = 2.5 \end{aligned}$$

$$\begin{aligned} \text{\#WBC count} &= N_{\underline{o}} \text{ of cells counted} \times 20 \times 2.5 \\ &= N_{\underline{o}} \text{ of cells counted} \times 50 / \mu\text{l} \end{aligned}$$

#Normal range

$$\# 4,000\text{-}10,000 / \mu\text{l} [4\text{-}10 \times 10^3 / \mu\text{l}]$$

Interpretation

❑ Increase in

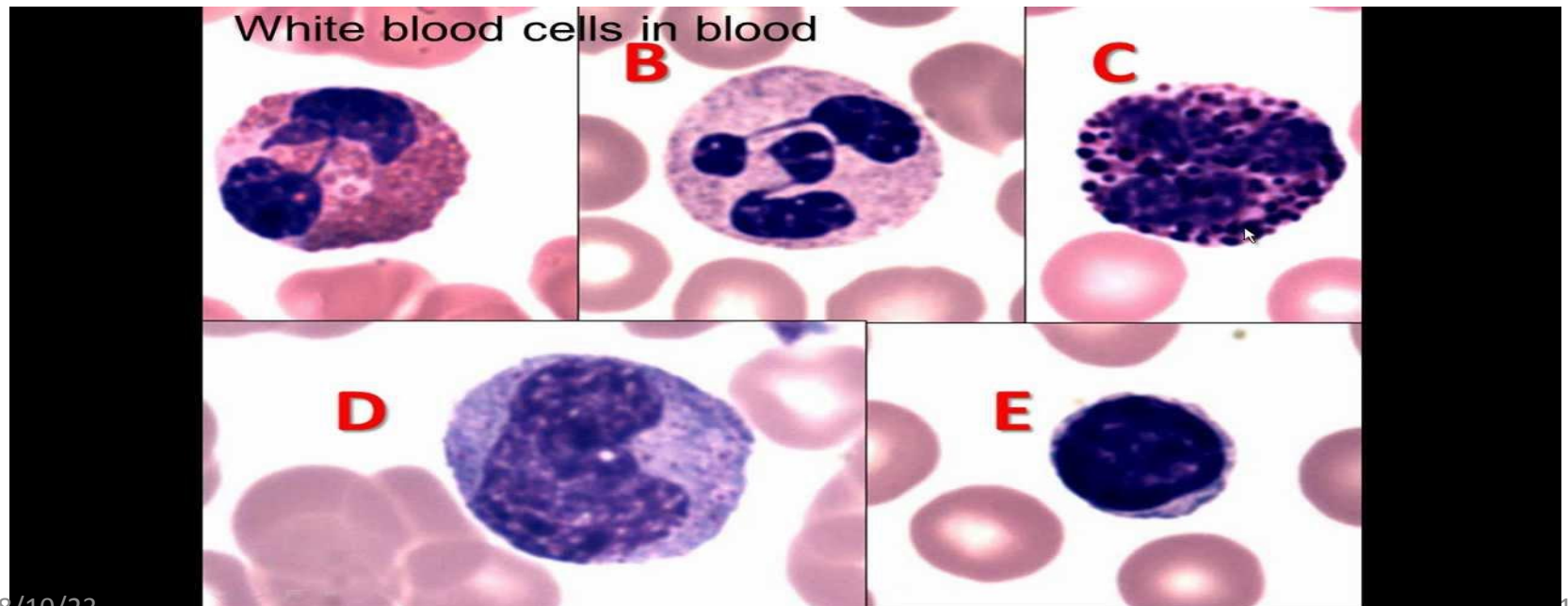
- ✓ Acute infection [Pneumonia, Tonsillitis]
- ✓ Inflammation & Tissue Necrosis
[Burn, Arthritis]
- ✓ Stress, Strenuous exercise
- ✓ Leukemia and Myeloproliferative disorders
- ✓ Metabolic disorders, Poisoning

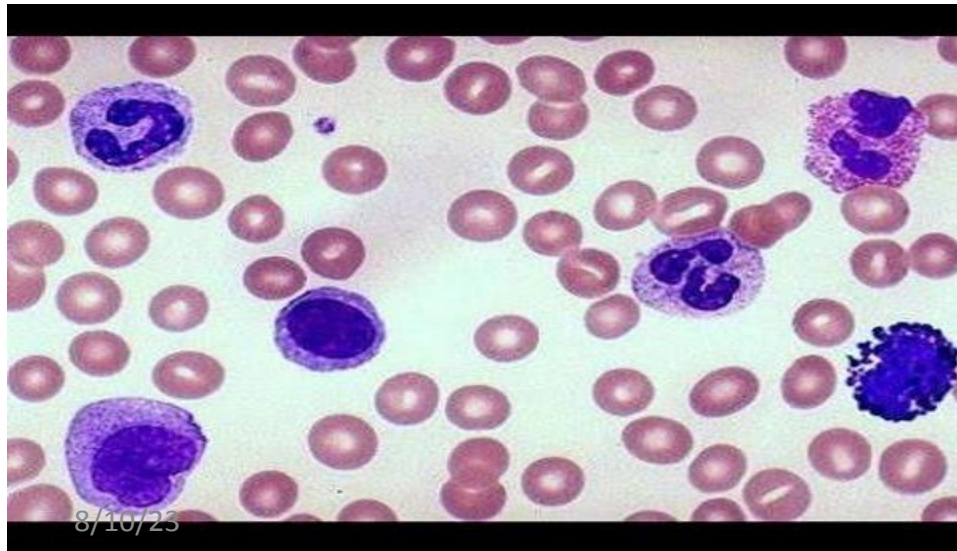
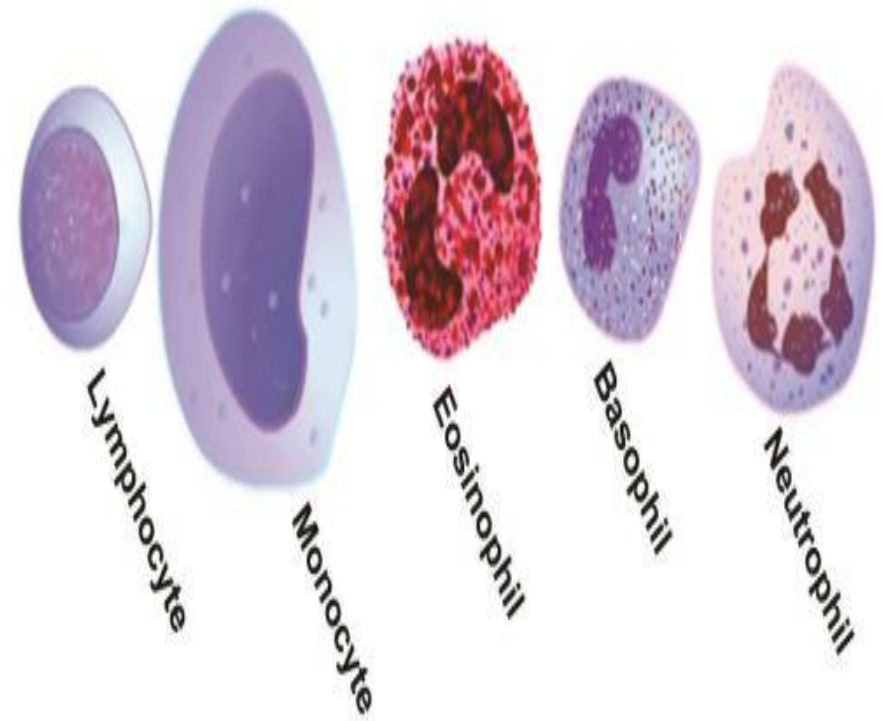
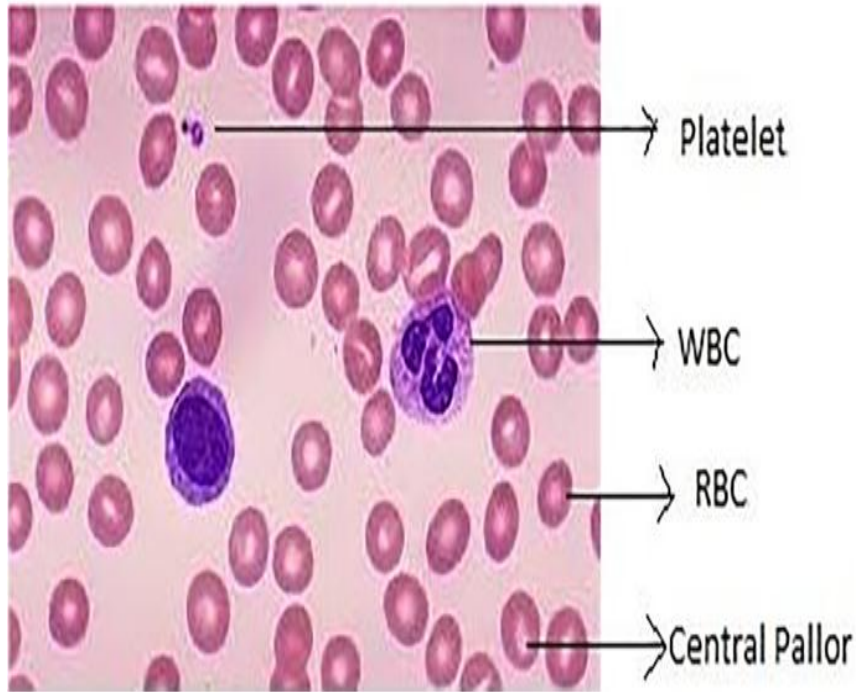
❑ Decrease in

- ✓ Viral infections [HIV]

2.Differential WBC count

- ❑ A differential WBC count provide information on the different white blood cells present in the circulating blood.
- ❑ White blood cells are grouped in to two
 1. **Granulocytes**→→Eosinophil, Neutrophil & Basophil
 2. **Agranulocytes**→→Monocyte & Lymphocyte





Cont'd

❑ How to identify each on the microscope?

- ✓ Color of **cytoplasm**
- ✓ Presence or absence of **granules**
- ✓ Size and color of **granules**
- ✓ **Size** of the cell
- ✓ Lobe or number of **nucleus**
- ✓ Shape of **nucleus**

Cont'd

☐ General staining steps [for diff count]

☐ Blood smear preparation

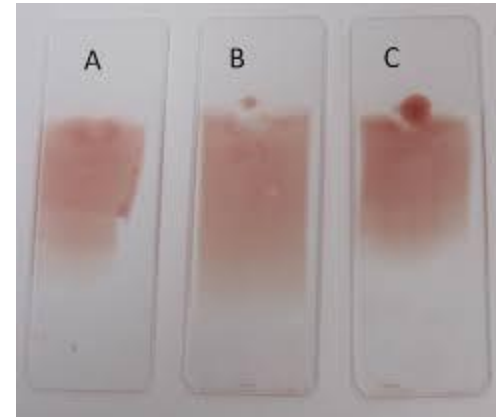
☐ Dry the smeared slide

☐ Staining

☐ Dry again

☐ Counting diff by Microscope

☐ Reporting



Morphology of different leukocytes under microscope

❑ Segmented neutrophil

- ✓ Size 12-15µm
- ✓ Lobed (segmented) nucleus with 3-5 lobes
- ✓ **Cytoplasm: Pale** staining with **small** neutrophilic granules

Segmented neutrophil



Cont'd

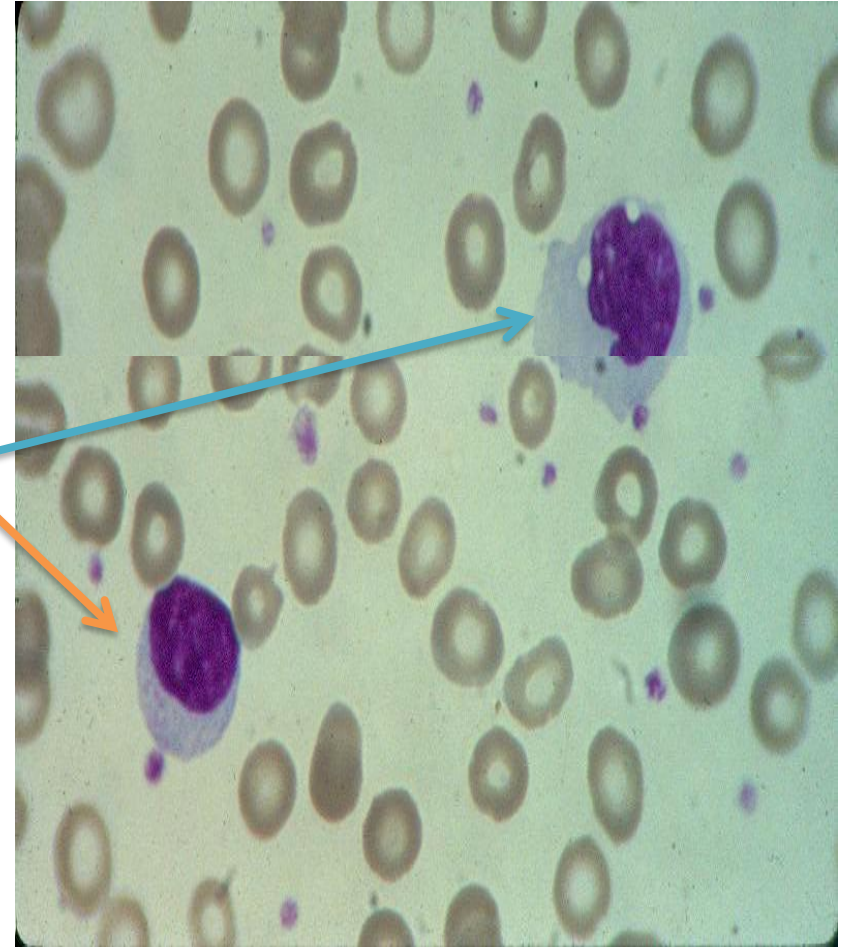
❑ Lymphocyte

❑ Small lymphocyte

- ✓ Size 10-12 μ m
- ✓ Dark round nucleus
- ✓ **Blue** cytoplasm

❑ Large lymphocyte

- ✓ Size 12-16 μ m
- ✓ Round , oval or indented nucleus

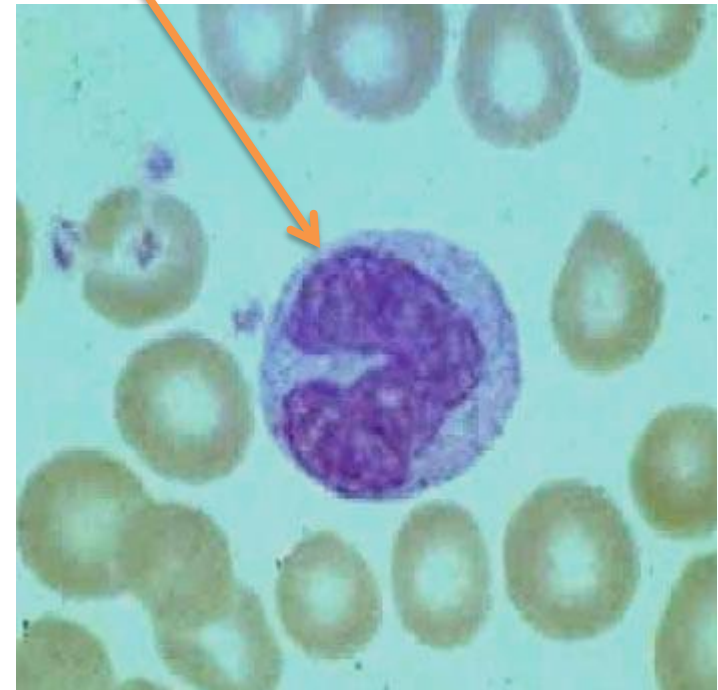


Cont'd

☐ Monocyte

- ✓ Size 15-20 μm
- ✓ Round, indented or folded nucleus

Monocyte

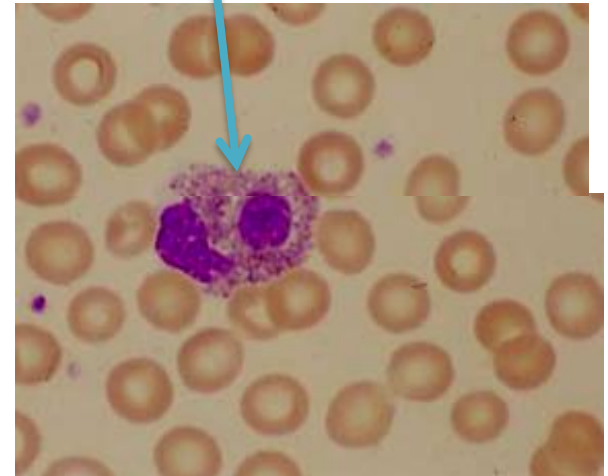


Cont'd

□ Eosinophil

- ✓ Size 12-17 μ m
- ✓ Bilobed nucleus
- ✓ **Pale** staining cytoplasm with **orange red** granules

Eosinophil

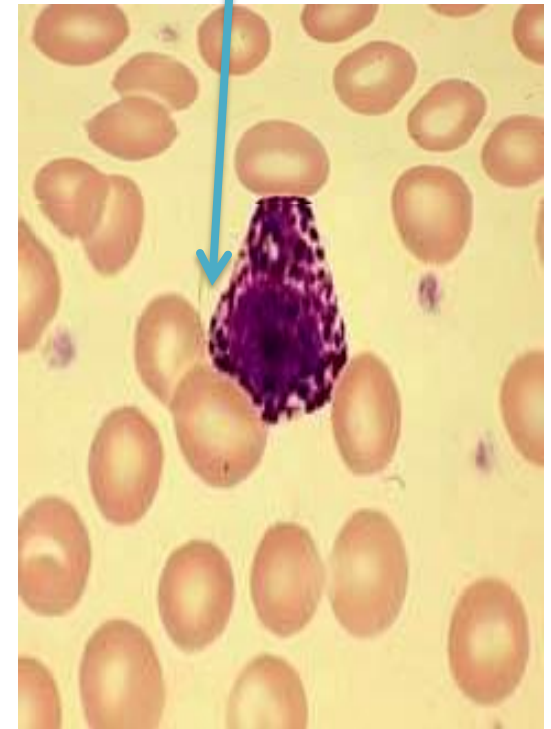


Cont'd

❑ Basophil

- ✓ Size 10-12µm
- ✓ Bilobed nucleus but usually obscured by granules
- ✓ Large **blue-violet** granules in the cytoplasm

Basophil



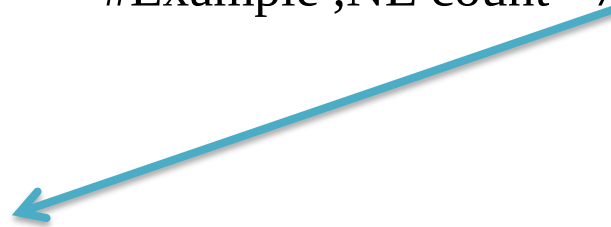
Interpretation

#Absolut count [AC]

- ✓ Multiplying the percentage [%] of the identified cells by **TWBC count**.
- ✓ **70% X TWBC count.**
- ✓ **If TWBC count=6000**
- ✓ **NE count=RC X 6000**
- ✓ **NE count =0.7 X 6000=4200 NEs/ μ L**

#Relative count [RC]

- ✓ Strictly the percent [%] counted in the WBC differential.
- ✓ #Example ,NE count =70%



Cont'd

Differential WBCs reference range

Neutrophils	$2.0 - 7.5 \times 10^9/l$ (40 – 75%)
Lymphocytes	$1.2 - 4.0 \times 10^9/l$ (21 – 40%)
Monocytes	$0.2 - 1.0 \times 10^9/l$ (2-10%)
Eosinophils	$0.02 - 0.6 \times 10^9/l$ (1-6%)
Basophils	$0.01 - 0.1 \times 10^9/l$ (0-1%)

Cont'd

❑ **Neutrophilia/Neutrophilic leukocytosis:**

- ❑ An increase in the N_o of circulating NE above normal ($>2.0-7.0 \times 10^9 /L$)
 - ✓ Physical and emotional stress
 - ✓ Infections
 - ✓ Tissue destruction or necrosis
 - ✓ Drugs and chemicals: lead, mercury
 - ✓ Metabolic disorders: uremia, diabetic acidosis

Cont'd

☐ **Neutropenia:**

☐ A reduction of the absolute NE count below $2.0 \times 10^9/L$

✓ Drugs (chloramphenicol, phenylbutazone)

✓ Ionizing radiation

✓ Myeloid hypoplasia

Cont'd

☐ **Eosinophilia:**

- ☐ An eosinophil count above $0.5 \times 10^9 /L$
 - ✓ Intestinal parasitic infections
 - ✓ Allergic diseases: bronchial asthma

☐ **Basophilia:**

- ☐ A basophil count above $0.2 \times 10^9 /L$
 - ☐ Rare condition
 - ✓ Occurs during: Allergic reactions, Chronic myelogenous leukemia

Cont'd

☐ **Monocytosis:**

- ☐ A monocyte count above $1.0 \times 10^9 / l$

- ☐ Occurs during

 - ✓ Monocytic leukemia

 - ✓ Recovery from acute infections, Tuberculosis,

☐ **Monocytopenia:**

- ☐ A monocyte count below $0.2 \times 10^9 / l$

- ☐ Occurs during

 - ✓ Treatment with prednisone, Leukemia

Cont'd

❑ Lymphocytosis

❑ $>4.0 \times 10^9 /L$

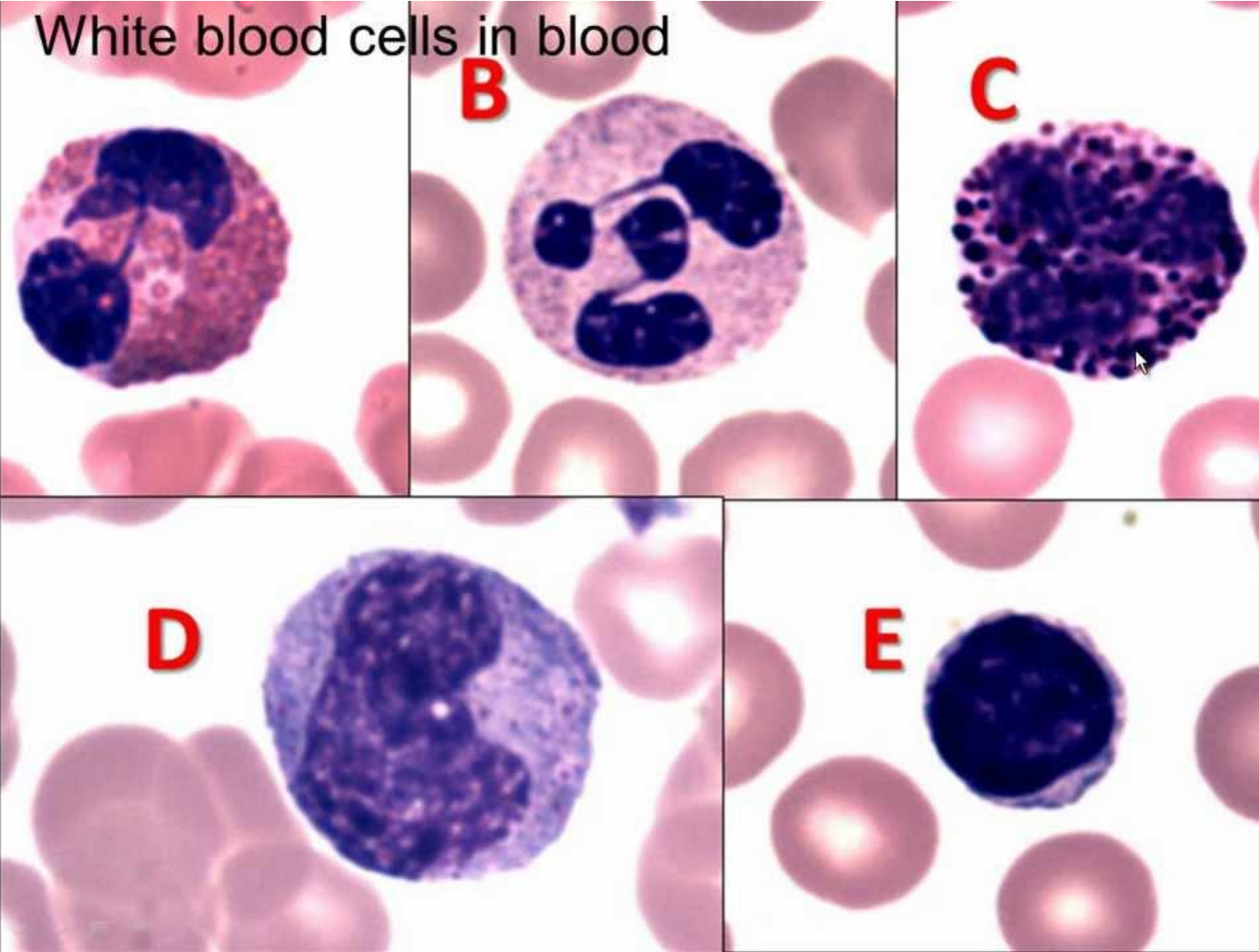
- ✓ Viral infections: Epstein-Barr virus ,cytomegalovirus
- ✓ Acute and chronic lymphocytic **leukemia**

❑ Lymphocytopenia

❑ $<1.0 \times 10^9 /l$

- ✓ Immune deficiency disorders: HIV/AIDS
- ✓ Drugs, radiation therapy

White blood cells in blood



3.Red Blood Cell Count [RBC]

❑ Significant of the test

- ❑ To know number of RBCs in physiological and pathological conditions.
- ✓ Used to diagnose and classify anemia
- ✓ To calculate red cell indices [RCI]

Principle of the test

- ❑ Whole blood is diluted 1:200 in [**Formal Citrate=Hayem's solution**] which maintains the disc like shape of RBC and prevents agglutination.
- ❑ After RBCs counted
 - ✓ The count is multiplied by **Dilution Factor** and **Volume Correction Factor**
 - ✓ The N_0 of RBCs are reported as number of cells per microliter (μl) of whole blood.

Specimen

- ☐ EDTA anticoagulated blood or
- ☐ Capillary blood can be used for counting RBC

Material and Reagents

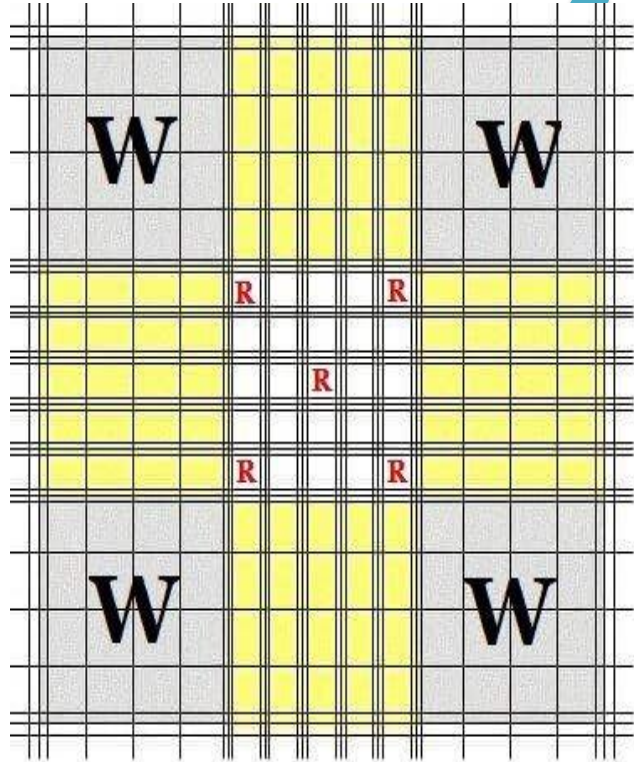
❑ Haemocytometer

- ✓ RBC Thoma Pippet → → Mark 101
- ✓ Cover Glass
- ✓ Diluting Fluid = **Hayem's** Solution = **Formal Citrate**
- ✓ Microscope
- ✓ Cotton
- ✓ Alcohol

Procedure

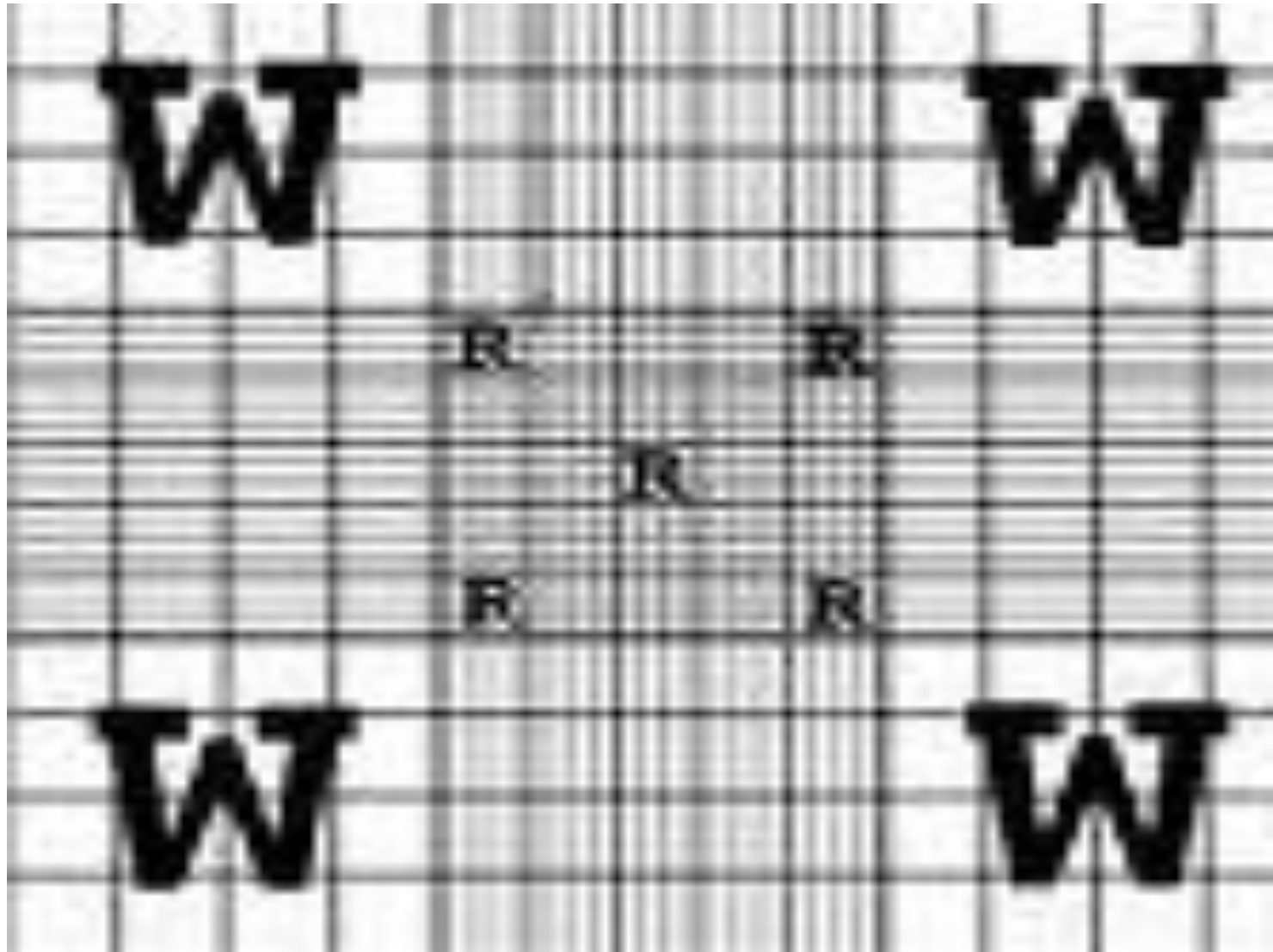
1. **Mix** the anticoagulated blood carefully
2. **Suck** the blood into pipette exactly up to **0.5** mark.
3. **Wipe** the excess blood outside the pipette using cotton
4. Take diluting fluid up to **101** mark
5. Gently rotate the pipette for **mixing**
6. **Discard** the first drops of unmixed **Hayem's solution** present at the stem of the pipette.
7. **Charge** the counting chamber and allow 2 minutes for settling of cells
8. Count RBCs in the center of **5** small squares of the Haemocytometer using microscope

About the Haemocytometer



- ❑ RBCs count in the central large square (1mmx1mm)
- The central large square is divided into 25 small squares
- Each small square is $1/5 \times 1/5 = 1/25$ (0.04mm^2) area and its volume is 0.004mm^3
- So volume used for RBC count is $5 \times 0.004\text{mm}^3 =$
 0.02mm^3 .

RBCs counting areas (R_1 - R_5)



Calculation and Result

$$\underline{1\mu\text{L}=\text{mm}^3}$$

$$\text{\#RBC Count} = N_{\underline{0}} \text{ of cells counted} \times \text{DCF} \times \text{VCF}$$

$$\text{DCF} = 200 (1:\textcolor{red}{200})$$

$$\text{VCF} = \text{Volume desired/Volume used}$$

$$= 1\text{mm}^3/0.02\text{mm}^3=\textcolor{red}{50}$$

$$\text{\#RBC count} = N_{\underline{0}} \text{ of cells counted} \times 200 \times 50$$

$$= N_{\underline{0}} \text{ of cells counted} \times 10,000/\mu\text{l}$$

Normal range

#Adult male : $5-6 \times 10^6/\mu\text{l}$

#Adult female : $4- 5 \times 10^6/\mu\text{l}$

Interpretation

☐ RBC count Increase in

- ✓ Polycythemia

☐ RBC count Decrease in

- ✓ Anemia
- ✓ Sever falciparum parasitic infection
- ✓ Blood loss

4.Methods of Hemoglobin Measurement /Quantification/

- ❑ Hemoglobin means blood globin or blood protein.
- ✓ It is often abbreviated as Hb or Hgb
- ✓ And is made up of Heme and Globin.
- ✓ Heme is a pigment which contains iron and it makes up about 4% of the hemoglobin molecule.

Hemoglobin Synthesis

- ❑ Heme is ring form, iron in the center, it mediates reversible binding of oxygen by hemoglobin.
- ❑ Globin is a colorless protein that surrounds and protects the heme molecule.
 - ✓ It makes up about 96% of the hemoglobin molecule.
- ❑ Hemoglobin synthesis requires the coordinated production of heme and globin.

Methods of Hemoglobin Measurement

- ❑ Is the determination of concentration of Hgb in red cells (whole blood).
- ✓ Hgb is reported in g/dL.
- ✓ The procedures for the Hgb determination are many and varied.
- ✓ These procedures can be grouped as:
 - 1. Specific gravity 3. Chemical
 - 2. Gasometric 4. Colorimetric Methods

Colorimetric Methods

Further grouped as:

1. Spectrometric methods

- ✓ Cyanmethemoglobin Method
- ✓ HemoCue® photometer
- ✓ Oxyhemoglobin Method.

2. Direct Matching Methods [Visual comparison]

- ✓ Acid Hematin Methods [Sahli-Hellige Method]
- ✓ Alkali Hematin Method
- ✓ WHO color scale

1. Sahli-Hellige (Acid hematin method)

Principle of the Procedure

- Blood is mixed with a 1% solution of HCL acid this breaks down the RBCs and converts the Hgb to a **dark brown** solution of **acid hematin**.
- The acid hematin solution is then → diluted with water until it matches the color of a standard on the Sahli-Hellige hemoglobinometer.

Necessary Reagents and Equipment

- ✓ Sterile blood lancet, 70% alcohol, and cotton or gauze.
- ✓ A pipette for measuring 20 cumm of blood (a Sahli pipette)
- ✓ A rubber sucking tube (aspirator)
- ✓ 1% HCl
- ✓ Distilled water with dropper.
- ✓ Glass stirring rod and standard Sahli-Hellige hemoglobinometer with
- ✓ Graduated tube.

Procedure

1. Fill the graduated tube of the hemoglobinometer to the **10** mark with the 1% HCL.
2. Make a finger puncture and draw blood into the **sahli pipette** to the **20** cumm mark.
3. With a piece of cotton, wipe the blood off the outside of the pipette.
4. Expel the blood into the graduated tube of the Sahli-Hellige hemoglobinometer.
5. Rinse the inside of the **sahli pipette** several times .

Cont'd

6. Now add 1 drop of the distilled water to the graduated tube and then mix with the glass stirring rod.
7. Continue the above process of adding 1 drop of water and stirring until the color of the solution in the **graduated tube** matches the color of the **glass standard** on the hemoglobinometer.
8. The hemoglobin reading is **the height the solution** has raised in the graduated tube.
9. Take the hemoglobin reading in -----g/dl

Disadvantage

- Fading of the color glass standard and
- Difficulty in matching it to the acid hematin solution.
- Acid Hematin is unstable and begins to fade almost immediately after it reaches its peak.

Cont'd

- Difficulty in matching the acid hematin with the glass standard
- Color matching is subjective/personal bias
- Interpersonal difference in reading the endpoint of dilution.
- As a result the Sahli-Hellige method is not as such recommended for Hb determination.

5.Packed cell volume (PCV=Hct) measurement

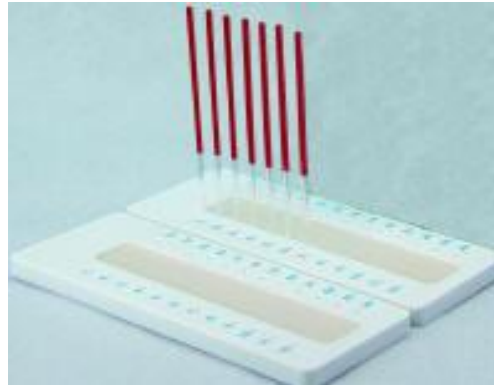
- ❑ Is a measure of the ratio of the volume occupied by the red cells to the volume of whole blood.
- ✓ The ratio is measured after appropriate centrifugation.
- ✓ Expressed as a decimal fraction or as percentage (%)

Materials

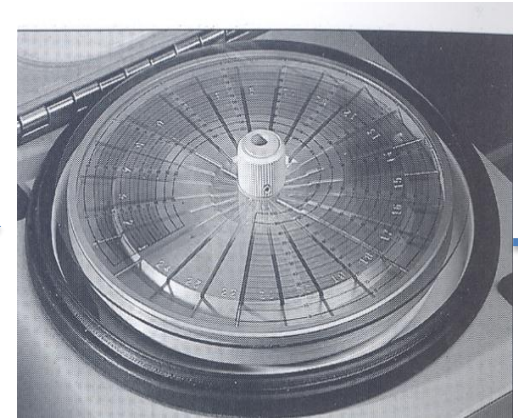
1. Sealant



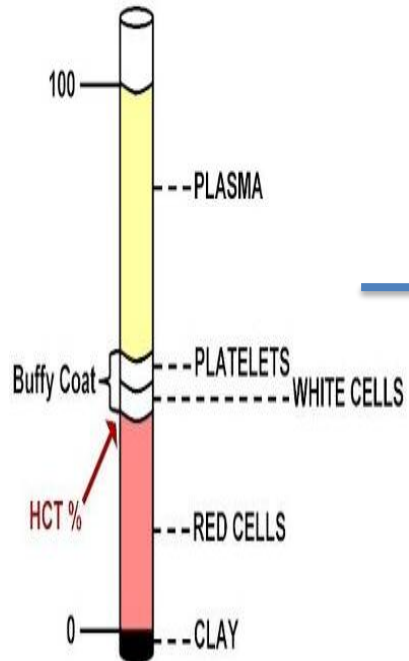
2. Capillary tube and sealant



3. HCT centrifuge



4. After centrifugation



5. HCT reader

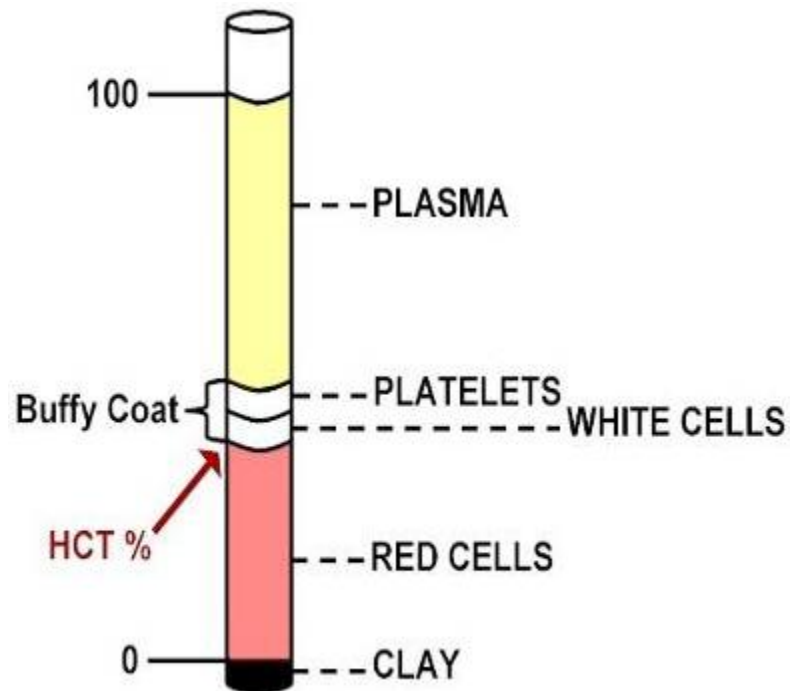


6. Normal range

- ❑ Adult Male 41– 55%
- ❑ Adult Female 35 – 48%
- ❑ Normal range varies among age, gender and altitude.

Cont'd

☐ After centrifugation



Significance of the test

- ✓ Enables the calculation of the red cell indices that are widely used in the classification of anemias.
- ✓ For screening of anemia
- ✓ To diagnose polycythemia and to monitor its treatment.

Hematocrit Determination

Methods

1. Macro method

- ✓ Wintrobe

2. Micro methods

- ✓ Adams microhematocrit method

1. Microhematocrit Method

Principle of test

- ❑ An Anti-coagulated blood in a **glass capillary tube** centrifuged in a microhematocrit centrifuge
- ✓ At 10,000-15,000 RPM for 5 minutes to obtain complete packing of the red cells
- ✓ The PCV value is read from the scale of microhematocrit reader OR
- ✓ Calculated by dividing the height of the red cell column by the height of the total column of blood (mm).

Specimen:

- ❑ Either well mixed EDTA anti-coagulated blood or capillary blood collected into a heparinized capillary tube

Equipment

- ❑ Microhematocrit **centrifuge** fixed speed (10,000-15,000 rpm)
- ❑ Microhematocrit **reader**, **Sealant** (plastic , clay)
- ❑ Heparinized Capillary **tubes** (red band).

Procedure

1. Fill About ($\frac{3}{4}$ th) of the heparinized capillary tube.
2. Seal the unfilled end using a sealant material.
3. Carefully locate the filled capillary in one of the numbered slots of the microhematocrit rotor **with the sealed end against the rim** gasket.
4. Centrifuge for 5 minutes at 10,000-15,000 RPM,

Cont'd

5.Immediately after centrifuging, read the PCV OR

6. $PCV = \text{Length of red cell column (mm)} / \text{Length of total column (mm)}$.

❑ Examples:if RBC=9mm & Whole blood 49mm

$$\checkmark PCV = 9\text{mm} / 49\text{mm} = 0.388 = 38.8\%$$

Interpretation

❑ Ethiopians: **Adult Males 41.6 – 55.1%**

Adult Females 35.3 – 48.8%

❑ Reference range varies between age, gender and altitude.

PCV

- ✓ Decreased in anemia
- ✓ Increased in polycythemia

Point

- ❑ The solutions for RBC manual counting (including Hayem's solution) are used to preserve the integrity of cellular membrane and it have no effect on leukocytes that are not eliminate.
- ❑ The effect of the presence of leukocytes on RBC counts is deleted by the dilution factor that is equal to 1:100 or more. Because this fact the presence of leukocytes on the Thomas is negligible.

6. Red Cell Indices

The red cell indices:

- ❑ Are absolute values calculated from the measured Hgb, RBC count and HCT value.
- ❑ They have considerable clinical importance in the diagnosis and classification of anemias.
- ❑ The red cell indices include
 1. Mean cell volume (MCV)
 2. Mean cell hemoglobin (MCH)
 3. Mean cell hemoglobin concentration (MCHC)

1.The Mean Cell Volume (MCV)

- ❑ The MCV provides information on **red cell size**.
- ❑ It measures **the average volume of a red blood cell** .
- ❑ It is measured in femtolitres (fL= 10^{-15} L) and is determined from the PCV and RBC count. It can be calculated as
- ❑ $MCV = \text{PCV} / \text{RBC count} / L$

Example: PCV = 0.45

$$RBC = 5 \times 10^{12} / L$$

$$MCV = 90 \text{ fL}$$

Interpretation

Normal Values

Men and Women: 80-100 fL .

☐ MCV

☐ Increased in

- Macrocytic anemias.. $MCV > 100 \text{ fl}$.

☐ Decreased in... $< 80 \text{ fl}$

- Iron deficiency anemia
- Microcytic anemia

2.The Mean Cell Hemoglobin (MCH)

- ❑ The average weight [amount] of hemoglobin in a red blood cell
- ❑ OR the amount of haemoglobin in picograms (pg) in an average red cell.
- ✓ It is calculated from the Hgb and RBC count:
- ✓ It is given by:

$$\text{MCH (Pg)} = \text{Hgb (g/L)} / \text{RBC count}$$

- ❑ Example: Hgb conc. = 150g/l

$$\text{RBC} = 5 \times 10^{12}/\text{L}$$

$$\text{MCH (pg)} = 30\text{pg}$$

Interpretation

❑ MCH is increased in

✓ Macrocytic anemia

❑ MCH is decreased in

✓ Microcytic anemia

✓ Iron deficiency anemia

❑ **Normal Value: Men and women: 27-31 pg**

3.The Mean Cell Hemoglobin Concentration

- ❑ It measures the average concentration [color] of Hgb in a red blood cell.
- ❑ OR the average hemoglobin per unit volume of red cells.
- ✓ $MCHC (g/l) = Hb (g/L)/PCV (\%)$

Calculate the MCHC: Hgb =13.5 g/dL=135g/l

Hct= 40.0%

Answer:

$$MCHC = 33.75\%[g\%]=337.5g/l$$

Interpretation

- ✓ **Normal Values: Men and women: 32-36 g% (320-360g/L)=32-36g/dl.**
- ✓ MCHC is increased in some cases of hereditary spherocytosis
- ✓ and decreased in iron deficiency anemia
- ✓ Cells with a normal concentration of Hgb are called normochromic.
- ✓ Cells with a lower normal concentration called **hypochromic**.

7.Erythrocyte Sedimentation Rate (ESR)

- ❑ ESR is the **rate of fall** (sedimentation) of red cells when blood is allowed to stand vertically undisturbed for
- ❑ a specified period of time, usually 1 hour.
 - ✓ The rate is expressed in mm/hr.

Significance of Measuring the ESR

- ❑ ESR is a non-specific test.
 - ✓ It increase in a wide range of infection bcz of change in plasma proteins, like increases in fibrinogen.
 - ✓ And many other factors including anemia, pregnancy.

Principle

- ❑ The ESR is determined **by filling** a narrow pipette with well mixed anticoagulated blood.
- ✓ And **standing** it in a vertical position for 1 hour.
- ✓ After 1Hr **the interface between the** plasma and the sedimented red cells is recorded in **mm/hour**.

Determination of ESR

❑ There are 2 basic methods

✓ Westergren and Wintrobe methods

Equipment

- ✓ Westergren pipette measure in [mm]
- ✓ A rubber teat or pipette filler
- ✓ Westergren rack (stand)
- ✓ Timer
- ✓ Anticoagulated blood [most tri-sodium citrate]

Specimen

❑ Sodium citrate anticoagulated blood



Procedure

1. Draw blood to the 0 mark of the Westergren pipette
2. And stand the ESR pipette vertically on the **ESR stand** under RT and avoid vibrations.
3. Set the timer for 1 hour.
4. After exactly 1 hour, read the level at which the plasma meets the RBCS in mm.

Reference range

Reference range (it depends on age,sex,books)

- ✓ Men 0mm to 10 mm/hour
- ✓ Women 0mm to 15 mm/hour

Cause of ESR raise and reduced

Raised ESR

- ✓ Anemia due to any cause
- ✓ Acute and chronic inflammatory conditions and infections, like:
 - HIV - Tuberculosis
 - Acute viral hepatitis & pregnancy

Reduced ESR

- In polycythaemia,
- Conditions associated with haemoconcentration
- Abnormal shaped RBCS as in sickle cell disease

8.Osmotic fragility of RBCs

9. Blood group